

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Probability and conditional probability

01

8 answers · Rationale-rich · Teach from doc

Q1**0.7**

Accept also: 7/10

The correct answer is $\frac{35}{50}$. Based on the table, there are a total of 50 tulips, and 35 of these tulips are from site A. The probability of selecting at random a flower from site A, given that the flower is a tulip, is equal to the number of tulips from site A divided by the total number of tulips, which can be written as $\frac{35}{50}$, or $\frac{7}{10}$. Note that 35/50, 7/10, and .7 are examples of ways to enter a correct answer.

Q2**42**

The correct answer is 42. It's given that in Nigeria on September 30, 2015, the probability of selecting an MTN subscriber from all Internet subscribers is 0.43, that there were p million, or $p(1,000,000)$, MTN subscribers, and that there were 97 million, or 97,000,000, Internet subscribers. The probability of selecting an MTN subscriber from all Internet subscribers can be found by dividing the number of MTN subscribers by the total number of Internet subscribers. Therefore, the equation $\frac{p(1,000,000)}{97,000,000} = 0.43$ can be used to solve for p . Dividing 1,000,000 from the numerator and denominator of the expression on the left-hand side yields $\frac{p}{97} = 0.43$. Multiplying both sides of this equation by 97 yields $p = (0.43)(97) = 41.71$, which, to the nearest integer, is 42.

Q3**C**

C. 13.3%

Choice C is correct. The sum of the number of states with a state-level sales tax is $39 + 6 = 45$. Of these states, 6 don't have a state-level income tax. Therefore, $\frac{6}{45} = 0.1333\dots$, or about 13.3%, of states with a state-level sales tax don't have a state-level income tax.

Choice A is incorrect. This is the number of states that have a state-level sales tax and no state-level income tax. Choice B is incorrect. This is the percent of states that have a state-level sales tax and no state-level income tax. Choice D is incorrect. This is the percent of states that have no state-level income tax.

Q4**.3833**

Accept also: 23/60

The correct answer is $\frac{23}{60}$. It's given that one of the participants will be selected at random. The probability of selecting a participant from group A given that the participant is at least 10 years of age is the number of participants in group A who are at least 10 years of age divided by the total number of participants who are at least 10 years of age. The table shows that in group A, there are 14 participants who are 10–19 years of age and 9 participants who are 20 + years of age. Therefore, there are $14 + 9$, or 23, participants in group A who are at least 10 years of age. The table also shows that there are a total of 30 participants who are 10–19 years of age and 30 participants who are 20 + years of age. Therefore, there are a total of $30 + 30$, or 60, participants who are at least 10 years of age. It follows that the probability of selecting a participant from group A given that the participant is at least 10 years of age is $\frac{23}{60}$. Note that 23/60, .3833, and 0.383 are examples of ways to enter a correct answer.

Q5

B

B. $\frac{5}{12}$

Choice B is correct. Since the participant will be selected at random, the probability of selecting a participant from group A, given that the participant is at least 10 years of age, is equal to the number of participants from group A who are at least 10 years of age divided by the total number of participants who are at least 10 years of age. Based on the table, in group A, there are 17 participants who are 10–19 years of age and 8 participants who are 20+ years of age. Therefore, there are a total of $17 + 8$, or 25, participants in group A who are at least 10 years of age. Based on the table, of the total number of participants, there are 30 participants who are 10–19 years of age and 30 participants who are 20+ years of age. Therefore, a total of $30 + 30$, or 60, of the participants are at least 10 years of age. Thus, the probability of selecting a participant from group A, given that the participant is at least 10 years of age, is $\frac{25}{60}$, or $\frac{5}{12}$.

Choice A is incorrect. This is the number of participants from group A who are at least 10 years of age divided by the total number of participants, rather than divided by the number of participants who are at least 10 years of age.

Choice C is incorrect. This is the probability of randomly selecting a participant from group A, given that the participant is 10–19 years of age, rather than given that the participant is at least 10 years of age.

Choice D is incorrect. This is the probability of randomly selecting a participant who is at least 10 years of age, given that the participant is in group A.

Q6**A****A.** $\frac{5}{32}$

Choice A is correct. It's given that set K consists of all the positive integers that are less than or equal to 160, where none of the integers are repeated. Therefore, set K consists of the integers 1 through 160, which is 160 numbers. Of these, the integers 1 through 50 are less than or equal to 50. Therefore, there are 50 numbers that belong to set K that are less than or equal to 50. Of these, half are even. Therefore, $\frac{50}{2}$, or 25, numbers that belong to set K are even and less than or equal to 50. The probability of selecting at random a number from set K that is even and less than or equal to 50 is equal to the number of numbers that belong to set K that are even and less than or equal to 50, which is 25, divided by the total number of numbers that belong to set K, which is 160. It follows that the probability of selecting at random from set K a number that is even and less than or equal to 50 is $\frac{25}{160}$, or $\frac{5}{32}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the probability of selecting at random a number from set K that is less than or equal to 50, not a number that is even and less than or equal to 50.

Q7**8**

The correct answer is 8. In this group, $\frac{1}{9}$ of the people who are rhesus negative have blood type B. The total number of people who are rhesus negative in the group is $7+2+1+x$, and there are 2 people who are rhesus negative with blood type B. Therefore, $\frac{2}{(7+2+1+x)} = \frac{1}{9}$. Combining like terms on the left-hand side of the equation yields $\frac{2}{(10+x)} = \frac{1}{9}$. Multiplying both sides of this equation by 9 yields $\frac{18}{(10+x)} = 1$, and multiplying both sides of this equation by $(10+x)$ yields $18 = 10+x$. Subtracting 10 from both sides of this equation yields $8 = x$.

Q8**C****C.** $\frac{15}{31}$

Choice C is correct. If a tree from one of these rows is selected at random, the probability of selecting a maple tree, given that the tree is 20 feet or taller, is equal to the number of maple trees that are 20 feet or taller divided by the total number of trees that are 20 feet or taller. It's given that there are 6 rows of birch trees, and each row of birch trees has 8 trees that are 20 feet or taller. This means that there are a total of 6×8 , or 48, birch trees that are 20 feet or taller. It's given that there are 5 rows of maple trees, and each row of maple trees has 9 trees that are 20 feet or taller. This means that there are a total of 5×9 , or 45, maple trees that are 20 feet or taller. It follows that there are a total of $48 + 45$, or 93, trees that are 20 feet or taller. Therefore, the probability of selecting a maple tree, given that the tree is 20 feet or taller, is $\frac{45}{93}$, or $\frac{15}{31}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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